

1. Introduction: From Free Will to Sense of Agency

The problem of free will has run through the history of Western thought like a red thread: are we, as individuals, the authors of our actions, or are they determined by external causes? Psychology does not aim to solve this philosophical problem, but it investigates a related concept that is far more empirically accessible: the sense of agency (SoA). This is defined as the subjective experience of being the initiator and controller of one's own actions and of their effects on the environment (Haggard & Tsakiris, 2009). Three fundamental aspects underpin SoA: intention (the desired state that one pursues through action), action (awareness of one's own actions), and effect (perception of the consequences of action) (Wen & Imamizu, 2022). These three components are directly linked to the motor, perceptual, and sensorimotor systems, each playing a major role in generating SoA.

It is important to emphasize from the outset that SoA does not refer to a global attitude towards life, but to specific experiences tied to isolated events or particular action sequences. Most experiments in this field focus on such micro-events. In the laboratory, SoA is typically studied by introducing a causal relationship between the participant's action and a sensory effect, in which the experimenter distorts either the temporal or the spatial relationship between the two (Farrer et al., 2003; Metcalfe & Greene, 2007; Sato & Yasuda, 2005; Moore et al., 2009). The participant must then judge whether he or she caused the observed effect. This type of design allows for fine manipulation of SoA and its systematic measurement.

SoA is not limited to one's own body. For example, when a person uses buttons to control an avatar in a video game, although the avatar's detailed movements are generated by the computer, that person feels a sense of agency over the avatar because its movements match the buttons pressed. If, however, a computer error occurs, the person may temporarily lose SoA due to the sudden mismatch between the avatar's movement and the buttons pressed (Wen & Imamizu, 2022). This example illustrates the fundamental principle of SoA: the match between action and effect.

2. Distinguishing SoA from the Sense of Ownership

SoA must be distinguished from the sense of ownership (SoO), which refers to the awareness that one's own body and its parts belong to oneself (Braun et al., 2018; see also Gallagher, 2000; Tsakiris et al., 2007). Although the two are closely related, they can be dissociated: I can feel that my leg belongs to me (ownership) without feeling that I initiated its movement (absence of agency). Such dissociations occur both in everyday life — for example, in the case of tics or involuntary movements — and in clinical conditions such as Tourette syndrome, anarchic hand syndrome, or somatoparaphrenia (Tsakiris, 2010). Furthermore, in the rubber hand

illusion, participants can feel that the fake hand belongs to them (SoO) without feeling that they control it (SoA).

The importance of SoA lies in the fact that it allows us to monitor our own actions and to distinguish them from external causes. When we act, the brain constantly compares the predicted sensory consequences with the actual feedback; a mismatch generates an error signal that enables us to adjust our behaviour and reduces the sense of agency (Engbert et al., 2008; Frith, Blakemore & Wolpert, 2000). In addition, SoA plays a central role in social interaction, allowing us to discern whether an outcome was produced by us or by our partner.

3. Theoretical Models of SoA

Four major theoretical models have been proposed to explain the generation of SoA. These models are not mutually exclusive, but complementary: together they provide a comprehensive framework that spans from simple sensorimotor mechanisms to complex cognitive inferences.

3.1 The Comparator Model

The comparator model is the classical account of SoA. According to it, the motor control system uses five types of representations (Carruthers, 2010; Frith, Blakemore & Wolpert, 2000): (a) the goal state — the desired final position of the body, also called motor intention; (b) the motor commands, of which two copies are produced through the inverse model; the copy that remains central is called the corollary discharge or efference copy; (c) the predicted sensory consequences, generated by the forward model based on the efference copy; (d) the actual sensory consequences of the movement (reafference); (e) an estimation of body position, resulting from the continuous combination of predicted and actual information. SoA arises when the comparison between (c) and (d) indicates a match; when the two do not coincide, the event is perceived as external, and SoA is reduced or absent.

However, the comparator model has significant limitations. It applies mainly to simple motor actions, with immediate and predictable sensory consequences, and does not explain SoA for complex actions, with delayed effects or multiple possible consequences. Moreover, it can only explain the implicit, low-level component of agency (FoA), not the explicit judgment (JoA), which involves conceptual inferences and social context. In addition, the model requires case-by-case adjustments to explain problematic data: in some conditions, non-visual actual sensory consequences do not appear to play a role in generating SoA, while in others, a mismatch in the auditory modality is sufficient to reduce SoA, even if visual feedback is congruent (Carruthers, 2010). This observation suggests that different sensory modalities have different weights depending on context — which led to the development of the multifactorial weighting model of Synofzik and colleagues (2008b).

3.2 Wegner's Retrospective Theory

The second major theoretical model — and the one that complements the comparator model — is the retrospective (or postdictive) theory proposed by Wegner (2002). Unlike the comparator model, which explains SoA through the comparison of prediction with actual sensory feedback, the retrospective theory holds that SoA is not based on prediction, but on a post-hoc causal inference: agency is experienced when the brain infers that its own thoughts caused the action, based on contextual cues available after the outcome.

Wegner (2002) proposed that this causal inference relies on three principles that must be jointly satisfied for SoA to occur. The first is priority: the thought or intention must temporally precede the action. If the intention appears after the action, the causal link cannot be established. The second is exclusivity: there must be no other plausible causes of the action. If an external agent could have produced the action, self-attribution is reduced. The third is consistency: the observed outcome must match the initial intention. If the outcome differs from what was intended, the causal inference is weakened (Wegner, 2002; see also Wen & Imamizu, 2022). This logic mirrors causal inference in science: the cause must precede the effect, must be the only plausible cause, and must produce a consistent effect when repeated.

These three principles were demonstrated experimentally in Wegner's classic studies. In the "helping hands" experiment (Wegner, Sparrow & Winerman, 2004), participants stood with their arms at their sides while a confederate moved their arms underneath, so that the confederate's movements appeared to be the participant's. When participants heard instructions describing the confederate's movements, they reported stronger SoA over those movements — even though they did not execute them. In the "I-spy" experiment (Wegner & Wheatley, 1999), participants believed they were jointly controlling a cursor on a screen with a confederate; when the confederate forced the cursor to stop on an object, and the object's name had been mentioned 5 or 1 second earlier, participants reported stronger SoA over the stop — even though they had not controlled it. These studies demonstrate that SoA can be induced by contextual cues (priority, consistency), even in the absence of a real action.

However, the retrospective theory also has limitations. It does not explain how SoA arises for simple, automatised actions, in which there is no explicit conscious intention (e.g., involuntary postural adjustments). Nor does it explain why SoA is reduced under conditions in which sensorimotor prediction fails but the context is favorable (e.g., in the case of passive movements in Dubynin & Shishkin, 2017).

3.3 The Bayesian Integration Model

Both the comparator model and Wegner's retrospective theory partially explain SoA, but neither succeeds in integrating all experimental data. This mutual limitation led to the development of a

third model — the cue integration model or Bayesian model — which proposes that SoA does not depend on a single type of cue, but on the weighted integration of multiple sources of information, both prospective and retrospective (Moore & Haggard, 2010; Synofzik, Vosgerau & Newen, 2008b).

According to this model, the brain computes the probability that an event is caused by the self ($p(\text{self}|\text{outcome})$) based on a Bayesian rule: the probability is proportional to the product of the probability that the self produced that outcome ($p(\text{outcome}|\text{self})$) and the prior probability that the self is the agent ($p(\text{self})$), divided by the general probability of the outcome ($p(\text{outcome})$) (Moore & Haggard, 2010; Wen & Imamizu, 2022). In other words, SoA is not a binary process (present or absent), but a probabilistic continuum that depends on how likely it is that one's action caused the observed effect.

A crucial aspect of this model is that the weight of each type of cue depends on context — more precisely, on its perceived reliability. When sensorimotor cues are clear and consistent (e.g., precise and real-time visual feedback), they receive a higher weight. When these cues are weak, ambiguous, or inconsistent, the weight shifts toward contextual and cognitive cues (e.g., expectations, beliefs, social context). This flexibility explains why SoA can be maintained even under conditions in which one type of cue is disrupted (Moore & Haggard, 2010).

3.4 Predictive Processing

The fourth theoretical model — predictive processing — extends Bayesian integration to a hierarchical architecture of the brain. According to this framework, the brain is a system that continuously generates predictions about the hidden causes of sensory input and attempts to minimise prediction error — the discrepancy between prediction and actual feedback (Apps & Tsakiris, 2014; Friston, 2010). SoA, in this view, is not a separate module, but the result of integrating prediction errors at multiple hierarchical levels, from low-level sensorimotor levels to high-level cognitive levels.

A central concept in predictive processing is precision weighting. Not all prediction errors are treated equally: the brain assigns each source of information a precision (the inverse of its perceived variability), which determines how much that error matters in updating predictions. When a sensory signal is clear and consistent, its precision is high, and the associated prediction error carries greater weight. When the signal is ambiguous or inconsistent, its precision is low, and the error carries less weight (Apps & Tsakiris, 2014; Friston, 2010).

4. Components and Classifications of SoA

4.1 The Feeling of Agency (FoA) and the Judgment of Agency (JoA)

An important distinction in the conceptual delimitation of SoA is that between the feeling of agency (FoA) and the judgment of agency (JoA) (Gallagher, 2012; Synofzik, Vosgerau & Newen, 2008b; Bayne & Pacherie, 2007). According to this dual model, SoA is not a unitary phenomenon, but manifests at two distinct levels of processing.

FoA is an experience represented by an immediate, pre-reflective, and non-conceptual sensation, which does not require conscious reflection or deliberation. It arises during action preparation, from the sensorimotor prediction of effects, and is closely tied to immediate bodily signals (proprioception, efference copy). In other words, FoA is the raw sensation that "I am doing this" — a sensation that is not directly accessible to introspection and that manifests through behaviour, not through reflection. It answers the question: "do I feel that I am doing this?"

JoA, by contrast, is a high-level process — a reflective, conceptual, and controlled judgment that arises after the outcome. It relies on an interpretation about the authorship of a particular action, constructed from comparing prediction with actual feedback and integrating contextual information (Synofzik et al., 2008a). JoA involves concepts (causality, authorship, intention) and beliefs about the link between action and outcome. It answers the question: "do I believe that I did this?"

It is important to emphasise that the two levels occupy different positions in the hierarchy of cognitive processing, without interpreting this in terms of importance or quality of experience. FoA is defined mainly by sensory input and bodily signals, whereas JoA is closer to complex cognition, concepts, and inferences. Both levels are part of SoA, but they operate differently and can be influenced by different factors. The two aspects are not always aligned and can be empirically dissociated: in studies on passive movements, participants did not report FoA for movements executed by a mechanical device, but they did report JoA — the belief that the action could have been theirs — based on the experimental context (Dubynin & Shishkin, 2017).

4.2 Prospective and Retrospective Cues: Mapping onto the Experimental Design

A third classification of SoA concerns the temporal moment at which a certain type of information contributes to agency. We can distinguish between prospective cues, which appear before or during the action, and retrospective cues, which appear after the outcome. This distinction is directly relevant to the design of the present study, because it maps onto the four components of SoA that the experimental manipulation was designed to isolate.

Prospective cues are signals available before or during the action, which anticipate the outcome and can determine SoA even before the effect is observed. These include action fluency (how easily and naturally response selection and preparation unfold), intention clarity (how precisely the action goal is mentally represented), and motor preparation (the readiness state of the motor system). Chambon and Haggard (2012) demonstrated that action fluency predicts SoA

independently of actual motor performance: when response selection is fluent, participants report stronger SoA, even if the action outcome is identical. In a subsequent study, Chambon, Moore and Haggard (2015) showed that transcranial magnetic stimulation (TMS) over the supplementary motor area (SMA) disrupts intentional binding, suggesting that the SMA is involved in generating the prospective predictions that support SoA. In the present design, the prospective component is captured by initiation time — the latency between face onset and the participant's action — which reflects motor preparation, response selection fluency, and anticipation of the effect's timing.

Retrospective cues, by contrast, are signals available after the outcome, which contribute to SoA through outcome monitoring and action-effect matching. These include comparing the initial prediction with actual sensory feedback, integrating contextual information about causality, and evaluating consequences. In comparator model terminology, retrospective cues correspond to detecting the match or mismatch between the efference copy and actual feedback (Frith, Blakemore & Wolpert, 2000). In Wegner's (2002) terminology, they correspond to the consistency criterion — the extent to which the observed outcome matches the initial intention. In the present design, the retrospective component is captured by the AOC condition (consistent/inconsistent/control), which reflects the match or mismatch between the intended direction and the observed direction of the avatar's gaze. This component corresponds to Wegner's (2002) consistency criterion.

A robust SoA does not rely exclusively on one type of cue, but integrates both sources of information, with their relative weight depending on context (Moore & Haggard, 2010). When prospective cues are clear and fluent, they can support SoA even before the outcome appears. When prospective cues are weak or ambiguous, SoA becomes more dependent on retrospective cues — on the match between what was intended and what was observed. This complementarity explains why SoA can be maintained even under conditions in which one of the two types of cues is disrupted.

Beyond these two temporal components, two further components of SoA must be considered in the experimental design. The first is the sense of ownership (SoO) — the awareness that one's own body and its parts belong to oneself (Braun et al., 2018; Tsakiris et al., 2007). In the present design, SoO is present in all conditions, including the control condition, because the participant always executes a motor response (pressing a key, moving the mouse, or touching the screen). For this reason, SoO cannot explain differences between conditions: it is constant and does not vary with the AOC manipulation. The second is intentionality / volition — the mental representation of the action goal. In the present design, this component is directly manipulated by AOC: in the consistent condition, the intention is fulfilled; in the inconsistent condition, the

intention is violated; in the control condition, the intention is absent. It is closely related to the retrospective component, as both are captured by the AOC condition.

This decomposition has two important methodological implications for data analysis. First, ANOVA on the AOC condition captures the full effect of agency — both the intentional component (action-outcome matching) and the temporal prediction component. Second, ANCOVA with initiation time as a covariate removes the temporal prediction component, thereby isolating the effect of the retrospective component. This distinction is crucial for interpreting the results: if the AOC effect remains significant after controlling for initiation time, then the effect can be confidently attributed to the retrospective component of SoA, rather than to temporal prediction.

An important methodological issue concerns SoO and the control condition. One might attempt to isolate SoO by comparing the action conditions (consistent, inconsistent) with the control condition. However, this comparison does not work: even in the control condition, the participant executes a motor action (pressing a key to identify the target), so SoO is not zero. The action vs. control comparison conflates SoO with intentionality and temporal prediction. The only clean comparison for SoO is across experiments — Exp 1 (keypress, minimal motor involvement) < Exp 2 (mouse, moderate involvement) < Exp 3 (pointing, maximal involvement).

4.3 Intentional Action as a Core Component of SoA

Having examined the components of SoA and their mapping onto the experimental design, it is necessary to delve deeper into a construct that traverses all the theoretical models discussed: intentionality. Although the terms "intention" and "intentionality" are often used interchangeably, they designate distinct concepts. Intention refers to the mental state that precedes action, i.e. the goal or desired state that the individual aims to achieve through their action. On the other hand, intentionality, in the philosophical sense of the term, refers to the quality of an action being guided by intention. Namely, this is the relationship between the mental intention and the actual unfolding of the action (Searle, 1983; Pacherie, 2008). In the context of SoA, intentionality is crucial: it determines whether the action is experienced as "mine" or not. In the next paragraphs we will focus on the intention viewed in different models.

In the comparator model, intention is represented by the goal state, that is the first of the five components of the motor control system (Frith, Blakemore & Wolpert, 2000). The goal state specifies the desired final position of the body and is also called the motor intention (Pacherie, 2008). It is transformed into motor commands through the inverse model, and the efference copy of these commands generates the sensory prediction. Thus, intention is the starting point of the entire causal chain that leads to SoA: without a clear intention, there is no prediction, hence no comparison and as a conclusion, no agency.

Wegner (2002) proposed that SoA depends on three criteria that must be jointly satisfied: priority (the intention temporally precedes the action), exclusivity (there are no other plausible causes), and consistency (the observed outcome matches the intention). Among these, consistency is the criterion most directly relevant to the present work: it determines whether the observed outcome is attributed to the self or to an external cause. When the outcome matches the intention, the causal inference is confirmed; when it does not, it is weakened. Thus, intentionality is not just a merely precondition of action, but also a criterion for the retrospective evaluation of agency.

Finally, in the cue integration model, intention is captured by the prior probability that the self is the agent ($p(\text{self})$) and the probability that the self-produced a given outcome ($p(\text{outcome}|\text{self})$) (Moore & Haggard, 2010). The clearer and more precise the intention, the stronger the prior, and the more likely the agency inference. In this sense, intention functions as a prospective cue that modulates the interpretation of retrospective cues (action-outcome matching).

In the discussion about the AOC and the modulation of the intentionality, we see that, in the present design, AOC directly operationalises intentionality. In the consistent condition, the participant intends a direction (e.g., left), and the observed direction of the avatar's gaze matches this intention, hence intentionality is fulfilled. In the inconsistent condition, the participant intends a direction, but the observed direction is opposite, thus intentionality is violated. In the control condition, there is no action from the participant, hence no intention to cause a specific direction, which means that intentionality is absent. Thus, AOC manipulates not only action-outcome consistency, but also the quality of the experience of being the author of that action.

If intentionality is a central component of SoA, and SoA modulates attentional orienting (Wen & Imamizu, 2022), then intentionality should also modulate the GCE. This prediction is supported by the pilot study of Petkova and Janyan (2026), which directly demonstrated that AOC modulates the GCE: the GCE was significantly larger in the consistent condition (48 ms) compared to the inconsistent condition (23 ms), and absent in the control condition (17 ms, non-significant). This suggests that intentionality, operationalised through AOC, is not merely an epiphenomenon of action, but an active factor that shapes how another's gaze orients our attention.

By this exposition, we have established that intentionality is a central component of SoA and that AOC directly manipulates it. However, intentionality is not the only variable that can modulate the effect of AOC on the GCE. In what follows, we examine range of motion, i.e. the extent of movement involved in the action, as the second central experimental variable of this thesis.

4.4 Range of Motion and Motor Involvement

After we discussed the fact that intentionality is a central component of SoA and that AOC directly manipulates it, it is necessary to examine a second central experimental variable of this thesis: range of motion. This variable refers to the spatial and temporal extent of the motor action, i.e. how much the body moves, how long the movement trajectory is, and how much time the motor system is involved in executing the action. In the present design, range of motion varies along a gradient from keypress (pressing a button, where we have a minimal, discrete movement) to mouse movement (moving the cursor, which is a moderate, continuous movement) to touchscreen pointing (touching the screen, that is a maximal, direct, and embodied movement).

The relationship between motor involvement and SoA is complex and not unidirectional. On the one hand, one might assume that greater motor involvement goes into more movement, which will cause more effort, and hence more proprioceptive signals that would lead to stronger SoA. On the other hand, the literature suggests that the relationship is not so simple. Braun et al. (2018) showed that SoA and SoO are distinct constructs, and increased motor involvement may affect them differently. Furthermore, cue integration theory (Moore & Haggard, 2010) suggests that the reliability of cues matters more than their quantity: a simple but precise and fluent action (such as a keypress) may produce stronger SoA than a complex but noisy action (such as mouse movement or pointing).

The body is not merely a vehicle for action, but also a source of signals that contribute to SoA. Proprioceptive signals, tactile feedback, and muscular sensations provide information about the unfolding of the action that is integrated with sensorimotor predictions to generate SoA (Tsakiris et al., 2007; Apps & Tsakiris, 2014). The more the body is involved in the action, the more signals are available. However, these signals can also be noisy, i.e. proprioceptive and temporal variability of large movements may reduce the precision of prediction and, implicitly, may weaken SoA. This tension between the quantity and quality of signals is central to the Hypothesis 3 prediction.

An important distinction in the motor literature is the one between discrete and continuous actions. A keypress is a discrete action: it has a clear beginning and end, it is rapid, symbolic, and does not require complex spatial coordination. Mouse movement and pointing are continuous actions: they require a spatial trajectory, sustained visual-motor coordination, and a longer duration. In terms of cue integration theory, discrete actions provide precise temporal cues (the moment of pressing is clear), while continuous actions provide rich spatial cues (trajectory, direction, extent) but are noisier temporally.

A particular aspect of continuous actions is eye-hand coupling: during visually guided movements, gaze and hand (or cursor) are coupled. Johansson et al. (2001) and Neggers & Bekkering (2000) showed that, during reaching and object manipulation, gaze anticipates and guides hand movement. This relationship has also been quantified in complex cognitive tasks.

Chen, Anderson and Sohn (2001) measured the correlation between gaze dwell time and cursor dwell time in web browsing: the average correlation was $r = 0.58$, with more than 50% of pages showing correlations greater than 0.8. Furthermore, 84% of regions visited by the cursor were also visited by gaze, and the average gaze-cursor distance was ~ 90 pixels (approximately 1 degree of visual angle). Chen and Lim (2013) replicated this pattern in a debugging task, showing that the frequency and duration of cursor visits to target areas were comparable to those of gaze. These data provide quantitative evidence that gaze and cursor are coupled during visually guided movements, fact that is a central aspect for Experiment 2.

However, the coupling between gaze and cursor is not one of simple simultaneous tracking. Huang, White and Buscher (2012) showed that, in web search tasks, the cursor lags gaze by approximately 700 ms: the user looks at an element first, then moves the cursor toward it. Moreover, the inverse hypothesis — that gaze follows the cursor — was not supported by the data. This suggests that eye-hand coupling is an anticipatory process: gaze guides future action, rather than tracking it in real time. In Experiment 2, this temporal dynamic is relevant: the participant looks toward the selected area before moving the cursor, and the avatar's gaze moves in the same direction — but with a different temporal sequence. This nuance links Experiment 2 to the literature on predictive processing (Apps & Tsakiris, 2014) and on prospective cues (Chambon & Haggard, 2012).

The specific relevance of eye-hand coupling for Experiment 2 lies in the fact that the participant moves the cursor toward an area of the screen to select the avatar's gaze direction. Due to gaze-cursor coupling, the participant's gaze is directed toward the selected area, and the avatar's gaze moves in the same direction — creating a situation of self-other mirroring in which one's own behaviour (gaze) and the other's behaviour (avatar's gaze) are aligned. This situation is theoretically distinct from Experiment 1, where the keypress does not involve spatial coordination of gaze. In summary, this means that, in Experiment 2, the participant's gaze is already directed toward the selected area before the cursor reaches it, a form of gaze pre-orienting that may create a stronger prediction than the cursor movement itself.

Based on these considerations, Hypothesis 3 predicts that the modulation of the GCE by AOC will increase as range of motion increases — keypress (Exp 1) < mouse (Exp 2) < pointing (Exp 3). This prediction is based on the idea that greater motor involvement produces richer proprioceptive signals and potentially stronger SoA, which in turn could amplify the effect on the GCE. However, there is also an alternative prediction: if motor noise and temporal variability of large movements reduce prediction precision, then the AOC effect might be weakened in Experiments 2 and 3. This theoretical tension is explored empirically in the three experiments.

We have established that intentionality (section 4.3) and range of motion (section 4.4) are the two central experimental variables of this thesis. In what follows, we examine how SoA is

measured, both explicitly and implicitly, and why we choose explicit measures (Likert) for the present work.

5. Measurement of SoA

5.1 Explicit Measures

SoA can be measured in two distinct ways (Reddy, 2022). In explicit measures, participants verbally report whether they consider they have control over their actions or whether they (or someone else) are the causal agent of observed actions (Synofzik, Vosgerau & Newen, 2008a). Although verbal reports contain rich information, they can be influenced by subjective criteria, biases, or decision-making disturbances. For example, a liberal criterion can lead to more "yes" responses compared to conservative criterion, but the liberal criterion is not necessarily linked to a stronger SoA (Wen & Imamizu, 2022). For this reason, researchers have developed indirect (implicit) measures.

5.2 Implicit Measures and Intentional Binding

In implicit measures, researchers study how voluntary actions modify perceptual processes — such as temporal, visual, tactile, or auditory perception — without asking the participant to explicitly reflect on their agency (Haggard, Clark & Kalogeras, 2002; Moore & Obhi, 2012). The most used implicit measure is intentional binding (or temporal binding). It refers to the perceived compression of the temporal interval between a voluntary action and its sensory effect: when a person performs a voluntary action that produces a sensory effect, the interval between the two events is perceived as shorter than the objective interval.

More precisely, two temporal shifts occur simultaneously: (1) the action is perceived as occurring later than in reality (action binding), and (2) the effect is perceived as occurring earlier than in reality (effect binding). The two events thus approach each other in subjective time. Action binding and effect binding reflect different aspects of the intentional binding effect and are useful for separating anomalies in action perception from those in sensory feedback perception. For example, patients with Parkinson's disease show abnormal action binding, while their effect binding is like that of healthy participants (Wen & Imamizu, 2022). Importantly, the intentional binding effect is not observed when the action is involuntary — for example, when it is triggered by brain stimulation (Wen & Imamizu, 2022). This suggests that binding requires a volitional component, not just a simple temporal contingency.

The classical method for measuring intentional binding uses the Libet clock: a circular dial on which a dot rotates continuously. The participant must report the position of the dot at the moment they executed the action and at the moment they perceived the effect (Libet et al., 1983; Haggard et al., 2002). By comparing the reported position with the objective position, the

subjective shift is calculated. This measure is considered implicit because it does not ask the participant to explicitly reflect on their agency.

5.3 Dissociations Between Explicit and Implicit Measures

Although both types of measures are considered valid, they are not interchangeable. Wen, Yamashita, and Asama (2015) experimentally demonstrated that implicit and explicit measures can be dissociated: in a study that manipulated action-effect delay (0–1000 ms) and arousal (via stimulus colour), longer delay significantly reduced explicit agency ratings, but increased the intentional binding effect. Moreover, arousal amplified binding, but did not influence explicit ratings. This double dissociation suggests that the two types of measures reflect different aspects of SoA.

A similar dissociation was reported by Grynszpan, Sahaï et al. (2019) in a joint action study. Participants jointly manipulated two connected haptic devices, believing that their partner was always another person, although sometimes the partner was replaced by a robot. The results showed a striking dissociation: explicit agency judgments were higher with the robot partner, while intentional binding occurred only with the human partner.

A recent critical review of the literature on agency in joint action (Le Besnerais et al., 2024) confirms that implicit and explicit measures of SoA can diverge systematically. The authors emphasize that intentional binding and explicit agency judgments reflect different levels of processing and recommend using both types of measures for a complete characterization of SoA.

5.4 Critiques of Implicit Measures

Nevertheless, the validity of intentional binding as a purely implicit measure has recently been contested (Reddy, 2022). It has been argued that "implicit" reports are not perceptual effects generated by SoA, but judgment effects: under conditions of perceptual uncertainty, which are inherent in Libet clock tasks, participants rely on heuristics and response biases that are independent of the agentic state (see also Joordens et al., 2004, for the disappearance of the effect when temporal localisation is performed via mouse click instead of verbal report). In addition, many other factors, such as causal relations between events, attention, or arousal, can affect temporal perception and influence the binding effect (Wen & Imamizu, 2022).

Further evidence that binding is not a pure indicator of SoA comes from the study by Wen et al. (2015), who showed that arousal amplifies intentional binding without influencing explicit agency ratings. Moreover, it has been demonstrated that the intentional binding effect can occur even in the absence of intention: in a virtual reality study, simply passively viewing a virtual hand pressing a button to trigger a tone produced a binding effect similar to that in the condition where the tone was triggered by the participant's actual movements (Wen & Imamizu, 2022).

This critique has a direct implication for the present work: in a rapid colour-discrimination paradigm, with multiple trials and short temporal intervals, measuring intentional binding would be impracticable and ambiguously interpretable. For this reason, the thesis uses explicit measures (Likert), which allow direct comparison between conditions at the level of trial sets and are less susceptible to the methodological artefacts of implicit measures. However, the thesis also uses reaction times (RT) as an implicit, low-level indicator of agency-related processing, capturing the feeling of agency (FoA) rather than the judgment of agency (JoA).

6. Action-Effect Binding: The Mechanism Linking Action to SoA

A central concept linking the AOC manipulation to SoA is action-effect binding — the tendency to mentally associate actions with their anticipated sensory effects. According to ideomotor theory, actions are not mentally represented solely by their movements, but by the sensory effects they produce (Hommel, Müsseler, Aschersleben & Prinz, 2001). When a person learns to press a button to produce a sound, the mental representation of the action already includes the anticipated sound. This anticipated representation of the effect is activated before the action is executed and guides response selection and preparation.

Kühn, Brass and Haggard (2011) experimentally demonstrated that this action-effect link is directly relevant to SoA. In a series of studies, they manipulated the consistency between the participant's action and the observed effect, measuring both explicit SoA and intentional binding. The results showed that when the observed effect matched the anticipated effect, participants reported stronger SoA; when the effect was inconsistent with the anticipation, SoA was reduced. Moreover, this modulation was present even when participants had no explicit intention to produce a specific effect — suggesting that action-effect binding operates at an automatic, pre-reflective level.

Applied to the design of the present study, action-effect binding provides the cognitive mechanism that explains why the AOC manipulation works. In the consistent condition, the participant intends a direction, activates the anticipated representation of the effect, and the observed effect matches this anticipation — which confirms the action-effect link and supports SoA. In the inconsistent condition, the participant intends a direction, but the observed effect is opposite — which violates the anticipated link, generates a prediction error, and reduces SoA. In the control condition, there is no action from the participant, hence no anticipated representation of the effect — and SoA is absent or minimal.

Thus, action-effect binding is not merely a theoretical phenomenon, but the mechanism through which AOC manipulates SoA. It explains why the consistency between action and effect is crucial: because actions are mentally represented in terms of their effects, and the match between the anticipated and observed effect is the condition for SoA to arise.

7. Clinical Implications: Schizophrenia and Autism

Disturbances of SoA have been documented in various clinical conditions. In schizophrenia, patients may attribute their own actions to external agents or, conversely, may feel that they caused events over which they had no control — a phenomenon known as delusions of control (Frith, 2014; for a review, see Braun et al., 2018). A recent systematic review of studies investigating intentional binding (IB) in neuropsychiatric disorders (Moccia et al., 2024) offers a comprehensive perspective on how SoA is disrupted across clinical conditions. In schizophrenia, patients show a hyper-binding effect — an implicit over-attribution of the sensory consequences of movement to the self (Haggard et al., 2003; Voss et al., 2010). This over-attribution is consistent with neurocognitive theories that emphasize the role of impaired sensorimotor predictions in the genesis of positive symptoms (Moccia et al., 2024). In Parkinson's disease, hyper-binding appears only in patients taking dopaminergic medication or reporting impulsive-compulsive behaviors, suggesting that dopamine plays a critical role in modulating SoA (Moore, Schneider et al., 2010; Ricciardi et al., 2017). Conversely, in Tourette syndrome and functional movement disorders, IB is reduced — suggesting an under-attribution of agency (Kranick et al., 2013; Zapparoli et al., 2020).

Less studied, but equally relevant to the present work, is the possible link between SoA and autism spectrum disorder (ASD). Individuals with autism often show particularities in processing social signals and in integrating sensory information, which could also affect the sense of agency. Although the literature is still limited, a few studies have reported subtle differences in SoA in individuals with autism, especially regarding the integration of sensory and motor cues (Zalla & Sperduti, 2015). For example, individuals with autism may have difficulty discriminating between their own actions and those of others, especially when sensory feedback is ambiguous or delayed (Grynszpan, Nadel et al., 2012).

These clinical observations converge on a broader theoretical question: is SoA connected to theory of mind (ToM), i.e. the capacity to attribute mental states to others? Several lines of evidence suggest that it is. Both SoA and ToM require self–other distinction: the ability to differentiate between one's own actions and mental states and those of others (Gallese, 2003; Decety & Sommerville, 2003). At the neural level, the temporo-parietal junction (TPJ) is implicated in both SoA and ToM (Schilbach et al., 2012; Saxe et al. 2004). Furthermore, the capacity to control another's gaze, which is a form of social agency, involves both SoA and ToM (Stephenson et al., 2018). Autism offers a natural test of this link: individuals with autism show well-documented difficulties in ToM (Baron-Cohen, 1995) and subtle differences in SoA (Zalla & Sperduti, 2015). If SoA and ToM are connected, then these differences should be correlated — a prediction that has implications for understanding both basic mechanisms and clinical profiles. This thesis examines this link indirectly, by testing whether AOC, a manipulation of the consistency between action and

outcome, modulates the gaze cueing effect, which is itself a foundational component of joint attention and ToM.

8. Conclusion: SoA as a Multifaceted Construct

In this section, we have examined the concept of sense of agency (SoA) and its components, providing a theoretical framework for understanding how agency can modulate gaze cueing. We began with the definition of SoA as the subjective experience of being the initiator and controller of one's actions (Haggard & Tsakiris, 2009) and distinguished SoA from the sense of ownership (SoO), showing that the two can be dissociated (Braun et al., 2018; Tsakiris et al., 2007).

We then reviewed the four major theoretical models of SoA — the comparator model (Frith, Blakemore & Wolpert, 2000), Wegner's retrospective theory (2002), Bayesian integration (Moore & Haggard, 2010), and predictive processing (Apps & Tsakiris, 2014) — and argued that they are complementary, together providing a comprehensive framework for understanding SoA.

We then examined the main classifications of SoA: the distinction between the feeling of agency (FoA) and the judgment of agency (JoA) (Synofzik et al., 2008b); the distinction between prospective and retrospective cues (Chambon & Haggard, 2012); and the distinction between the different components of SoA — SoO, temporal prediction, intentionality, and action-outcome matching. We showed that these components map directly onto the design of the present study: RTs capture the implicit component (FoA), Likert ratings capture the explicit component (JoA), initiation time captures the prospective component, and the AOC condition captures the retrospective component.

We then examined the two ways of measuring SoA — explicit and implicit — and argued that they are not interchangeable, but reflect different aspects of agency (Wen et al., 2015; Grynszpan et al., 2019; Le Besnerais et al., 2024). Finally, we introduced the concept of action-effect binding (Hommel et al., 2001; Kühn et al., 2011) as the cognitive mechanism explaining why the AOC manipulation works.

Part IV — Does Agency Change Perception and Attention? And Can It Extend to Social Cues?

In the previous section, we examined the concept of sense of agency (SoA) and its components, providing a theoretical framework for understanding how agency can modulate gaze cueing. We established that SoA is a multifaceted construct that can be experimentally manipulated through the consistency between action and effect (AOC). But what happens after agency is generated? What are its consequences for perceptual and attentional processes? This section addresses these questions, moving from the antecedents of SoA to its consequences. We

will examine four lines of evidence showing that agency is not merely an epiphenomenon of action, but an active factor that shapes how we perceive and orient our attention. And as a conclusion, we will ask whether this effect extends to social stimuli such as gaze.

The first line of evidence comes from intentional binding (IB). As discussed in section 5.2, IB refers to the perceived compression of the temporal interval between a voluntary action and its sensory effect. Haggard, Clark and Kalogeras (2002) demonstrated that voluntary actions are perceived as shifted later in time, and their effects as shifted earlier — the two events "approach" each other in subjective time. This effect does not occur for involuntary movements, suggesting that IB reflects a process specific to agency. Moreover, IB is not merely a perceptual curiosity: it is considered the perceptual signature of agency (Haggard, 2017). When action and effect are causally linked, the brain integrates them temporally, and this integration can influence how subsequent events are processed. Thus, IB provides the first evidence that agency changes perception — in this case, the perception of time.

However, a crucial question arises: does IB coincide with the explicit sense of agency? If the two measures were unrelated, IB could not be considered a valid marker of agency. Imaizumi and Tanno (2019) addressed this question directly. In two experiments, participants pressed a key that triggered an auditory (Experiment 1) or visual (Experiment 2) outcome after variable delays (100–900 ms). In each trial, participants rated their sense of agency over the outcome and estimated the action–outcome interval. The results showed that longer delays decreased both agency ratings and intentional binding. Critically, the authors found intra-individual correlations between agency ratings and intentional binding on a trial-by-trial basis in both outcome modalities — that is, when a participant felt stronger agency on a given trial, they also showed stronger temporal binding on that same trial. This finding is important because it demonstrates that IB and explicit SoA are not dissociable artefacts, but two manifestations of the same underlying process. Moreover, it validates the use of IB as an implicit measure of agency in paradigms that manipulate action–outcome consistency, such as the AOC manipulation used in the present study.

But what determines the strength of IB? Beyond the mere presence of voluntary action, the degree of control over the outcome appears to be critical. Beck, Di Costa and Haggard (2017) manipulated the extent to which participants could control the intensity of a somatosensory outcome. In one condition, participants chose between two actions that differed in their probability of producing a high- or low-intensity outcome. In another condition, action selection was meaningless with respect to the outcome. The results showed that both action binding and outcome binding increased when participants had probabilistic control over the outcome. This finding demonstrates that intentional binding is sensitive not only to the presence of a voluntary

action, but also to the degree of control the agent has over the outcome — a key component of the AOC manipulation in the present study.

The second line of evidence comes from sensory attenuation. Blakemore, Frith and Wolpert (1999) showed that self-produced stimuli — such as the tactile sensation of one's own hand touching an object — are perceived as less intense than identical externally produced stimuli. This effect is interpreted because of sensorimotor prediction: the brain "subtracts" the anticipated sensory consequences of its own action, thereby reducing the neural response to self-produced stimuli (Blakemore et al., 2000; Frith, Blakemore & Wolpert, 2000). Sensory attenuation is not merely a perceptual curiosity: it shows that agency modifies the perceptual magnitude of events. If agency can reduce the intensity of tactile stimulus, it might also modulate the salience of a visual stimulus or a social cue. This line of evidence supports the idea that agency not only accompanies perception but also shapes it.

But why does sensory attenuation occur, and is it driven by motor action or by the sense of agency itself? Brown, Adams, Parees, Edwards and Friston (2013) provided a normative answer within the framework of active inference. According to their account, sensory attenuation is not an anomaly or a quirk of motor control, but a necessary consequence of the way action and perception are coupled. In active inference, movement is produced by descending proprioceptive predictions that are fulfilled by classical reflex arcs. However, for these predictions to prevail over sensory evidence, the precision of sensory prediction errors must be attenuated — otherwise, the sensory evidence that one is not moving would override the descending predictions, and movement would be precluded. Sensory attenuation is therefore the Bayes-optimal withdrawal of precise sensory evidence during movement, equivalent to "attending away" from the consequences of self-made acts.

This account has two important implications for the present work. First, it demonstrates that sensory attenuation and agency attribution are intimately linked: the same mechanism that allows movement to occur also determines whether a sensation is attributed to the self or to an external cause. When attenuation fails, like in the case of schizophrenia, the force-matching illusion disappears, and patients may develop delusions of control, falsely attributing their own actions to external agents (Brown et al., 2013; Shergill et al., 2005). This clinical evidence shows that agency is not merely a cognitive label attached to action, but a perceptual phenomenon grounded in the modulation of sensory precision. Second, it provides a unifying theoretical framework that connects the predictive processing model discussed in section 3.4 to the empirical phenomena of sensory attenuation and force-matching. If agency modulates perception through precision weighting, then it is plausible that the same mechanism modulates the salience of social cues — including gaze.

However, a crucial question remains: is sensory attenuation driven by motor action, by the sense of agency over the outcome, or by both? Han et colleagues (2021) addressed this question directly by dissociating motor action from sense of agency. In their paradigm, participants could control whether a sound would be presented by means of a willed inaction: if they chose not to press a button by a certain time, the sound was played; if they pressed, it was not. Thus, participants had complete control over the sound — and therefore a sense of agency over it — without performing any motor action. The results were striking: N1 suppression did not occur in the absence of a temporally coincident motor action, but Tb suppression did occur when participants had control over the sound. This dissociation demonstrates that sensory attenuation is not a unitary phenomenon. The N1 component appears to index motor-based prediction (requiring an efference copy), whereas the Tb component appears to index the sense of agency per se, independent of movement. This finding has two important implications. First, it confirms that agency can modulate perception independently of motor action — a stronger claim than the one supported by Brown et al. (2013), who linked attenuation primarily to movement. Second, it suggests that the effects of agency on perception are component-specific: different aspects of perceptual processing are influenced by different factors.

The third line of evidence comes from studies on action-based attention and eye-hand coupling. Research has shown that attention is automatically drawn to locations where our actions have an effect (Wen & Imamizu, 2022). For example, when a participant controls an object, their attention is preferentially directed toward that object, even if the task does not require it (Wen & Imamizu, 2022). Furthermore, during visually guided movements, gaze and hand (or cursor) are coupled: Johansson et al. (2001) and Neggers & Bekkering (2000) showed that gaze anticipates and guides hand movement, and Chen, Anderson and Sohn (2001) and Chen and Lim (2013) quantified this relationship in complex cognitive tasks. Huang, White and Buscher (2012) showed that, in web search, the cursor lags gaze by ~700 ms, suggesting that eye-hand coupling is an anticipatory process, not simultaneous tracking. These findings show that agency and attention are tightly coupled: our action not only changes what we perceive, but also where we orient our attention.

But is agency limited to objects and non-social events? The fourth line of evidence suggests it is not. Recent studies have shown that people can feel agency over social outcomes, including the behaviour of other agents. For example, in gaze-contingent paradigms, an avatar's or confederate's gaze is controlled by the participant's action: when the participant presses a button, the avatar's gaze moves in a specific direction. Participants report stronger SoA when the gaze is contingent on their action, compared to conditions in which the gaze is independent (Stephenson et al., 2018). Moreover, Grynszpan et al. (2019) showed that agency over a partner (human vs. robot) can be dissociated from implicit measures (IB), and Le Besnerais et al. (2024) confirmed that agency in joint action is a complex phenomenon that cannot be reduced to a

simple “we-agency”. These studies suggest that agency can extend beyond one's own body to other social agents — including their gaze.

Despite these advances, there is a major gap in literature. Most studies linking agency to gaze cueing ask: “Gaze → Agency” — that is, how gaze following or eye contact influences SoA (Ulloa et al., 2019; Stephenson et al., 2018). However, the reverse direction — “Agency → Gaze cueing” — has not been directly tested. The question is: if an observer causes another's gaze (and especially if that causation is consistent with their intention), does this modulate the magnitude of the gaze cueing effect? This is the central question of this thesis. To the best of our knowledge, no previous study has directly tested whether AOC over a gaze modulates the GCE. This thesis addresses this gap by experimentally testing whether action-outcome consistency (AOC) modulates the GCE in three experiments with increasing range of motion.

Rationale for the Current Study

Literature establishes three things. First, the gaze cueing effect (GCE) is flexible and can be modulated by facial, observer, and contextual factors (Part II). Second, the sense of agency (SoA) changes perception and attention for non-social stimuli: it compresses perceived time (intentional binding), attenuates the intensity of self-produced stimuli (sensory attenuation), and orients attention toward locations where action has an effect (action-based attention) (Part IV). Third, people can feel agency over social outcomes, including over another's gaze (Grynszpan et al., 2019; Stephenson et al., 2018).

What is missing is a direct test of whether agency, and in particular action-outcome consistency, modulates the GCE. Existing studies have manipulated either face characteristics, observer traits, or social context. None has manipulated the observer's own immediately preceding action as the source of the gaze. This thesis directly addresses this gap.

Action-outcome consistency (AOC) is the central operational manipulation of the thesis. It determines whether intentionality is fulfilled, violated, or absent. In the consistent condition, the avatar's gaze direction matches the participant's intended direction; in the inconsistent condition, it is opposite; in the control condition, there is no action from the participant, hence no intention to cause a specific direction. Thus, AOC directly operationalises Wegner's (2002) consistency criterion and manipulates the retrospective component of SoA (see section 4.3).

To verify whether the AOC manipulation indeed produces differences in SoA, the thesis uses explicit Likert ratings. Participants answer three questions after each set of trials: (1) Did I predict the avatar's gaze direction? (2) Did I cause the avatar's gaze direction? (3) Did I control the avatar's gaze direction? These questions target distinct dimensions of SoA, namely prediction, causality, and control, and allow a direct comparison between AOC conditions. Although the

design does not allow a direct trial-by-trial causal test, the ratings provide an explicit validation of the manipulation.

In all three experiments, the participant sees an avatar whose gaze direction is determined by their action. The main task is a rapid colour discrimination on a target that appears congruent or incongruent with the avatar's gaze direction. The AOC manipulation has three levels: consistent, inconsistent, and control. Range of motion varies across experiments: keypress (Experiment 1), mouse movement (Experiment 2), and touchscreen pointing (Experiment 3). In Experiment 1, the participant presses a button; in Experiment 2, they move the cursor; in Experiment 3, they touch the screen directly. In all experiments, initiation time is recorded as a measure of the prospective component.

H1: The gaze cueing effect (RT incongruent – RT congruent) will be largest in the consistent condition, intermediate in the control condition, and smallest (or absent/reversed) in the inconsistent condition.

This prediction is based on the pilot study by Petkova and Janyan (2026), which showed a GCE of 48 ms in the consistent condition, 23 ms in the inconsistent condition, and 17 ms (not significant) in the control condition. If H1 is confirmed, it would demonstrate that agency, which is operationalised through AOC, modulates attentional orienting in response to gaze.

H2: SoA ratings will be highest in consistent condition, intermediate in control condition, and lowest in the inconsistent condition.

This prediction follows directly from the definition of AOC: consistency between action and outcome should produce the strongest SoA, inconsistency the weakest, and absence of action an intermediate or reduced SoA. H2 extends the pilot study, which did not measure SoA explicitly.

H3: The modulation of the GCE by AOC will increase as range of motion increases, namely keypress (Exp 1) < mouse (Exp 2) < pointing (Exp 3).

This prediction is based on the idea that greater motor involvement produces richer proprioceptive signals and potentially stronger SoA, which in turn could amplify the effect on the GCE. However, there is also an alternative prediction: if motor noise and temporal variability of large movements reduce prediction precision, then the AOC effect might be weakened in Experiments 2 and 3. This theoretical tension is explored empirically.

The three experiments test this hypothesis along a gradient of motor embodiment. Together, they provide a systematic test of how agency — and in particular action-outcome consistency — modulates the gaze cueing effect. In what follows, Chapter 2 describes the method, results, and discussion of each experiment.

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